STUDY OF DIFFERENT TYPES OF NETWORK CABLES

Ex No: 2
23/7/26

AIM:

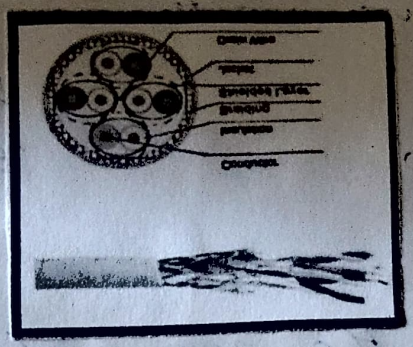
To understand different types of Network cables.

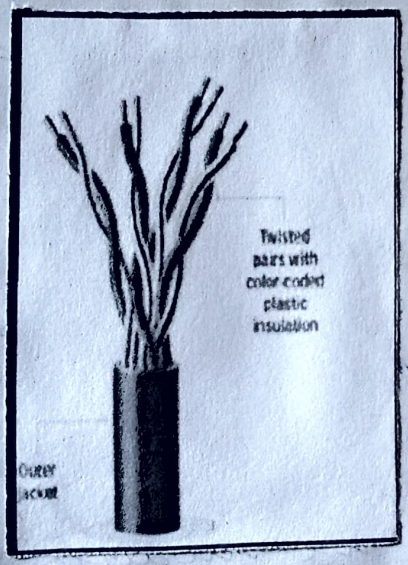
Cable Type	Category	Max data transmission	Advantages/ disadvantages	Application / use
UTP	category 3	10bps	**Advantages:** ★ cheaper in cost ★ Easy to install as they have a smaller overall diameter **Disadvantages** ★ More prone to (EMI) Electromagnetic interference and noise	10 Base-T Ethernet
	Category 5	Upto 100 Mbps		Fast ethernet, Gigabit Ethernet
	Category 5e	1 GPs		Fast ethernet, Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	**Advantages:** ★ shielded ★ Faster than UTP ★ Less susceptible to noise and interference. **Disadvantage** ★ Expensive ★ Greater installation effort	Gigabit Ethernet, 10 G Ethernet (55m) widely used in data centres
SSTP	Category 7	10 Gbps		Gigabit Ethernet 10 G Ethernet (100m)
Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	**Advantages:** ★ High bandwidth ★ Immune to interference ★ Low Loss bandwidth ★ Versatile **Disadvantage:** ★ Limited distance ★ cost ★ size is bulky	Speed of signal is 500m. Television network High speed internet connections.

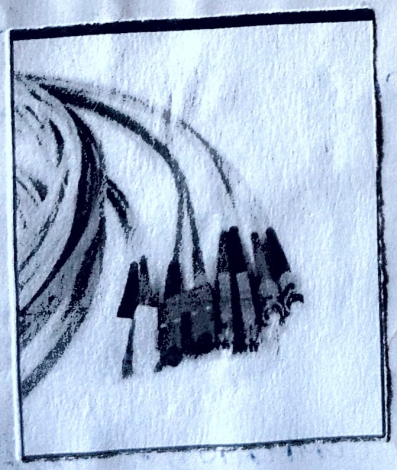
Cable Type	Category	Maximum data transmission	Advantages / Disadvantages	Applications / user
Fibre Optics Cable	Single mode Multi mode	100 Gbps	Advantages: * High speed * High bandwidth * High security * Long distance Disadvantages: * Expensive * Required skilled installers	Maximum distance of fibre optical cable is around 100 m

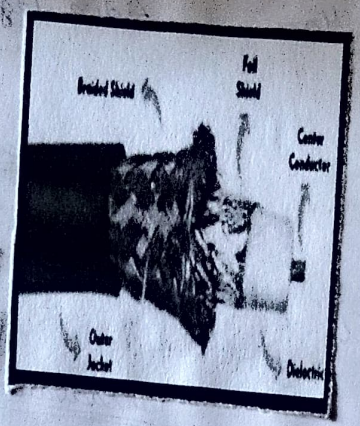
RESULT:

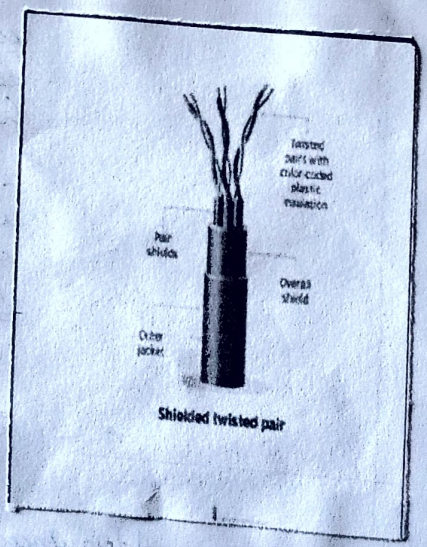
Thus, we have learned the different types of network cables for computer Networks.




Twisted pairs with color-coded plastic insulation
Outer jacket







Shielded twisted pair